# 谭晓锋 (Xiaofeng Tan)

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广东深圳

## 研究方向

RL, AIGC, and world models, with a focus on human-centric perception, generation, and understanding.

## 教育经历

**东南大学，计算机科学与技术，硕士** 2024.09 – 至今

CSRankings（全球，2024–2026）：AI #23 · CV #57 · ARWU 2026 全球 #89

- **GPA:** 88.6/100；**实验室:** PALM 实验室，导师：王洪松 教授
- **荣誉:** 东南大学一等学业奖学金（前 10%）；腾讯-SEU 奖学金（前 2.5%）；招生夏令营排名第二

**深圳大学，计算机工学 & 数学理学（双学位，荣誉学位）** 2020.09 – 2024.06

CSRankings（全球，2024–2026）：AI #27 · CV #32 · ARWU 2026 全球 151–200

- **GPA:** 88.9/100 · 4.07/4.50
- **荣誉:** 优秀毕业生代表（1/300+），并于毕业典礼发言；作为优秀学生事迹入选深圳大学 2024 招生指南

## 发表论文

* indicates equal contribution.

♦ **When Policy Entropy Constraint Fails: Preserving Diversity in Flow-based RLHF via Perceptual Entropy**

Xiaofeng Tan, Jun Liu, Bin-Bin Gao, Yuanting Fan, Xi Jiang, Chengjie Wang, Hongsong Wang, Feng Zheng

**NeurIPS 2026** · **Spotlight** (292/30,709 submissions, 0.95%) CCF-A, CORE-A*

[arXiv] [Project]

♦ **MotionRFT: Unified Reinforcement Fine-Tuning for Text-to-Motion Generation**

**TPAMI 2026**

Xiaofeng Tan*, Wanjiang Weng*, Hongsong Wang, Fang Zhao, Xin Geng, Liang Wang CCF-A, SCI-Q1

*Extended from EasyTune (ICLR 2026).*

[arXiv] [Project] [Code]

**EasyTune:** Efficient Step-Aware Fine-Tuning for Diffusion-Based Motion Generation ICLR 2026

Xiaofeng Tan*, Wanjiang Weng*, Haodong Lei, Hongsong Wang CCF-A, CORE-A*

[PDF] [Project] [Openreview]

**ReAlign:** Text-to-Motion Generation via Step-Aware Reward-Guided Alignment AAAI 2026

Wanjiang Weng*, Xiaofeng Tan*, Junbo Wang, Guo-Sen Xie, Pan Zhou, Hongsong Wang CCF-A, CORE-A*

[PDF] [Project] [Code]

**SoPo:** Text-to-Motion Generation Using Semi-Online Preference Optimization NeurIPS 2025

Xiaofeng Tan, Hongsong Wang, Xin Geng, Pan Zhou CCF-A, CORE-A*

[Project] [PDF] [Code]

**FG-Diff:** Frequency-Guided Diffusion Model for Skeleton-Based Video Anomaly Detection TIP 2026

Xiaofeng Tan, Hongsong Wang, Xin Geng, Liang Wang CCF-A, SCI-Q1

[Project] [PDF] [Code]

**MGBOD:** Fuzzy Granule Density-Based Outlier Detection with Multi-Scale Granular Balls TKDE 2025

Can Gao (Supervisor), Xiaofeng Tan, Jie Zhou, Weiping Ding, Witold Pedrycz CCF-A, SCI-Q1

[PDF] [Code]

**ConsistentRFT:** Dual-Grained Policy Optimization for Hallucination Reduction in Flow-based RL

Xiaofeng Tan, Jun Liu, Yuanting Fan, Bin-Bin Gao, Xi Jiang, Xiaochen Chen, Jinlong Peng, Chengjie Wang, Hongsong Wang, Feng Zheng *Under Review*

[Project] [arXiv] [Code]

## 学术服务

**Conference Reviewer:** NeurIPS 2026 (**Top Reviewer**), ICML 2026 (**Gold Reviewer**), AAAI 2027, ICLR 2027
**Journal Reviewer:** Transactions on Machine Learning Research (TMLR), International Journal of Data Science and Analytics (IJDSA)

## 科研经历

**Tencent Inc.**, Research Intern @ HY — Mar 2026 – Present

*Mentor:* Jiaxin Lin

- Spatial reward modeling and curriculum-guided RL for text-to-image generation: tool-grounded spatial verification with SpatialCritic and rollouts progressing from foundational skills to complex compositions (Getting Space Right, under review; joint work with UC Merced).

**University of California, Merced** — Apr 2026 – Present

Remote Visiting Student @ VLLab — *Mentor:* Ming-Hsuan Yang

- Remote collaboration on spatial reasoning and reward reliability for image generation, through the joint Getting Space Right project with Tencent HY.

**Tencent Inc.**, Research Intern @ YouTu Lab — Jul 2025 – Mar 2026

*Mentor:* Jun Liu, Feng Zheng

- Analyzed why policy entropy can remain constant while perceptual diversity collapses in flow-based RLHF; developed perceptual-entropy regularization to improve the quality–diversity trade-off (PEC, NeurIPS'26 Spotlight).
- Investigated visual hallucinations caused by limited exploration and trajectory imitation; developed dynamic-granularity rollouts and consistent policy-gradient optimization (Consistent-RFT, under review).

**Singapore Management University**, Remote Visiting Student @ LV Lab — Jul 2024 – Mar 2025

*Mentor:* Pan Zhou

- Developed semi-online preference optimization that pairs online unpreferred motions with offline preferred examples, addressing offline overfitting and online sampling bias (SoPo, NeurIPS'25).
- Co-developed step-aware reward modeling and reward-guided sampling to improve text-motion alignment and motion realism during denoising (ReAlign, AAAI'26; co-first author).

**Southeast University**, Master's Student @ PALM Lab — Apr 2024 – Present

*Mentor:* Hongsong Wang

- Co-developed step-wise reward optimization and a unified reward model for heterogeneous motion representations (EasyTune, ICLR'26; MotionRFT, TPAMI'26; co-first author on both). On HumanML3D with MLD, EasyTune achieves FID 0.132 at 22.10 GB peak memory, using 15.22 GB less than DRaFT-50.
- Developed frequency-guided diffusion with perturbation training for skeleton-based video anomaly detection, improving robustness to unseen normal motion and emphasizing global-motion reconstruction (FG-Diff, TIP'26).

**Shenzhen University**, Undergraduate Research Assistant — Oct 2022 – Oct 2023

*Mentor:* Can Gao

- Developed multi-scale granular-ball and fuzzy-rough-set methods for detecting heterogeneous outliers (MGBOD, TKDE'25; first student author), and three-way decision-based multi-view co-detection (TWCOD, IJAR'23; first author).

## 荣誉奖项

**硕士阶段 / Master's Studies**

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| **ICML 金牌审稿人** | 国际机器学习大会 | 2026 |
| **NeurIPS Top Reviewer** | 神经信息处理系统大会 | 2026 |
| **腾讯-SEU 奖学金（前 2.5%）** | 东南大学（9,000 元） | 2025 |
| **一等学业奖学金（前 10%）** | 东南大学（12,000 元） | 2025 |

**本科阶段 / Undergraduate Studies**

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| **优秀毕业生代表** | 深圳大学（毕业典礼发言） | 2024 |
| **入选招生指南** | 深圳大学（优秀学生事迹） | 2024 |
| **荣誉学士学位（前 3%）** | 深圳大学 | 2024 |
| **优秀毕业生（前 5%）** | 深圳大学 | 2024 |
| **荔园之星（特等奖，前 0.73%）** | 深圳大学（20,000 元） | 2022, 2023 |
| **拔尖创新人才奖学金** | 广东省高考理科前 2%（20,000 元） | 2020 |